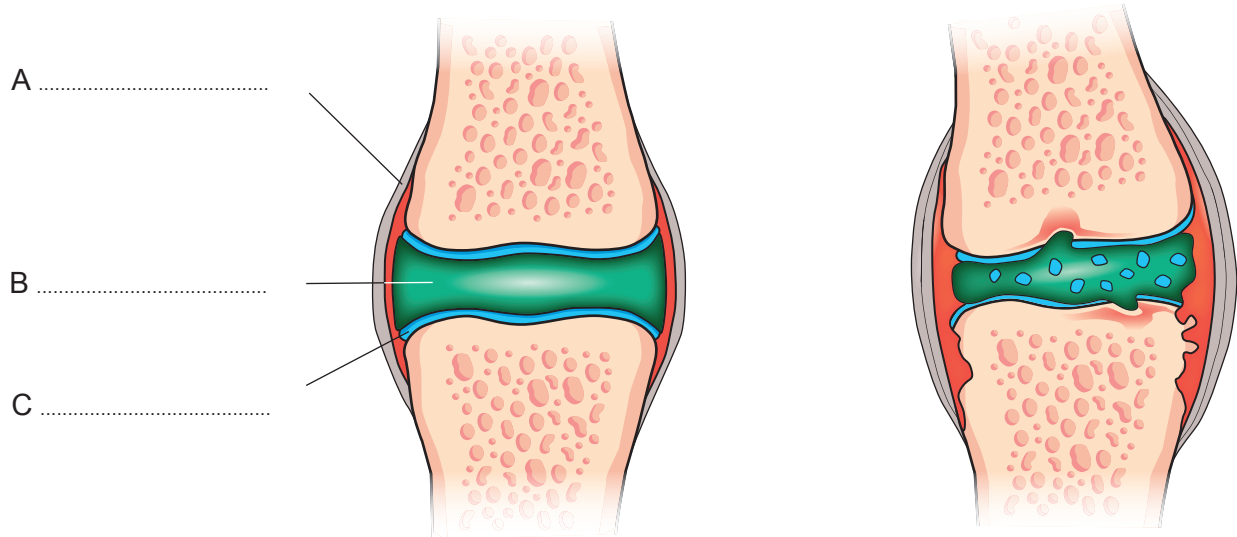


3. The upper and lower leg are connected by the knee joint.

The lower leg is connected to the foot by the ankle joint.

- (a) The knee is a synovial hinge joint.  
The diagram shows a healthy knee joint and one damaged by osteoarthritis.



- (i) Label the diagram using only the words from the box.

[3]

synovial fluid

ligament

muscle

cartilage

- (ii) I. State **one** cause of osteoarthritis.

[1]

.....

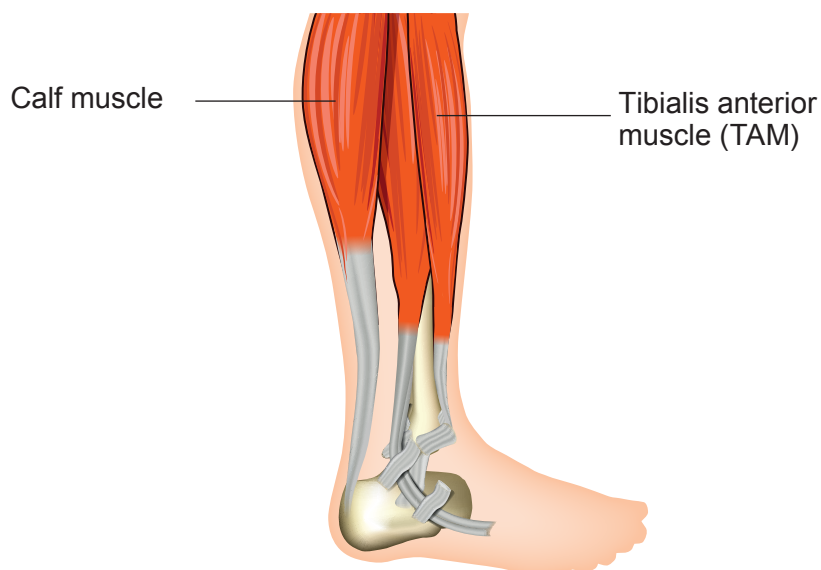
- II. State why a person suffering from osteoarthritis has knee pain.

[1]

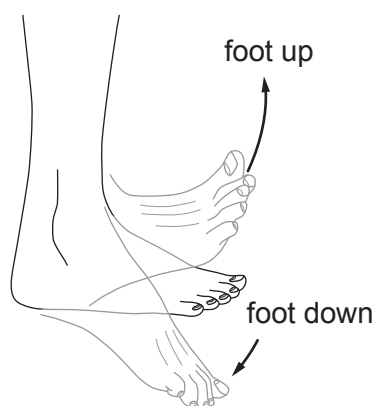
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- (b) The ankle joint is controlled by a pair of antagonistic muscles. These are the calf muscle and the tibialis anterior muscle (TAM).



Explain how these muscles work together to move the foot up and down as shown in the diagram. [2]



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